🚢 🚢 TITANIC SURVIVAL PREDICTION

🌟 CODSOFT DATA SCIENCE INTERNSHIP – TASK 1

🎯 OBJECTIVE

The objective of this project is to develop a Machine Learning model that predicts whether a passenger survived the Titanic disaster based on passenger information such as age, gender, passenger class, fare, and other details. This project helps in understanding data preprocessing, classification algorithms, and model evaluation techniques.

📁 PROJECT DESCRIPTION

The Titanic Survival Prediction project includes:

📊 Titanic dataset loading and analysis

🧹Data cleaning and handling missing values

🔢 Converting categorical data into numerical format

🤖Training a Machine Learning classification model

📈 Predicting passenger survival

🎯 Evaluating model accuracy

🛠️TOOLS USED

🐍 Python

💻 VS Code

📊 Pandas

🤖Scikit-learn

📈 NumPy

📊 FEATURES

1. Dataset Loading 📌

Loads Titanic passenger data from CSV files.

2. Data Cleaning 📌

Handles missing values and removes unnecessary columns.

3. Data Preprocessing 📌

Converts text values such as gender and embarkation ports into numerical values.

4. Model Training 📌

Uses Logistic Regression to train the prediction model.

5. Survival Prediction 📌

Predicts whether passengers survived or not.

6. Accuracy Evaluation 📌

Calculates model performance using accuracy score.

📚 CONCEPTS USED

✔ Data Cleaning

✔ Data Preprocessing

✔ Machine Learning

✔ Classification Algorithms

✔ Logistic Regression

✔ Train-Test Split

✔ Accuracy Evaluation

📈 KEY LEARNINGS

✔ Understanding Machine Learning workflows

✔ Working with real-world datasets

✔ Handling missing data effectively

✔ Training and evaluating classification models

✔ Improving analytical and problem-solving skills

💡 INSIGHTS

Machine Learning can predict outcomes based on historical data.

Data preprocessing is an important step before model training.

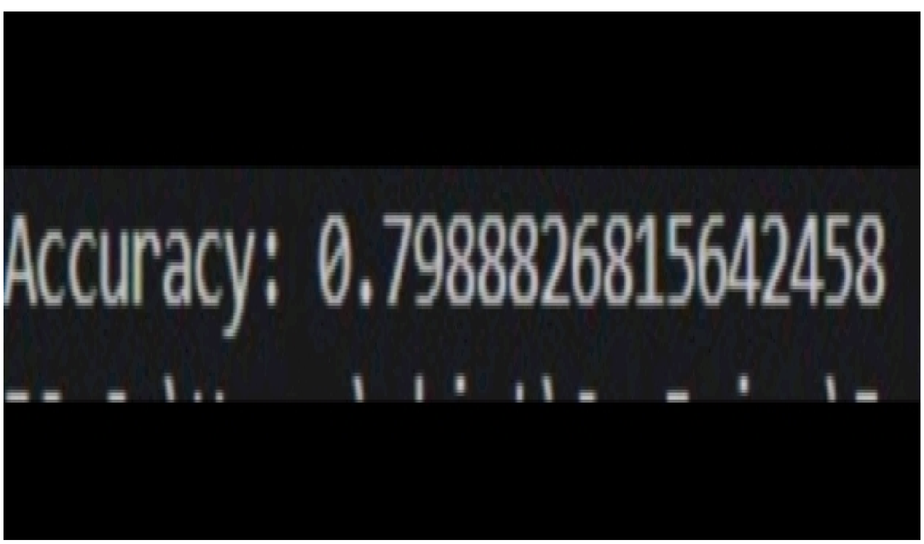
Logistic Regression is effective for binary classification problems.

Accurate predictions depend on data quality and feature selection.

🏁 CONCLUSION

The Titanic Survival Prediction project successfully demonstrates how Machine Learning can be used to predict passenger survival based on historical Titanic data. This project enhanced my understanding of data preprocessing, classification models, and performance evaluation as part of the CodSoft Data Science Internship.

📸OUTPUT



✔ Titanic Dataset Loaded Successfully

✔ Data Cleaning Completed

✔ Logistic Regression Model Trained

✔ Survival Predictions Generated

✔ Model Accuracy Achieved: 79.89%

👩 💻 AUTHOR

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